O número de permutações de n elementos

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Seja $\sigma^{(n)}\colon A_n \to A_n$, $A_n = \{a_1, a_2, \ldots, a_n\}$ e σ bijetora;

$$S_n \coloneqq \left\{\sigma^{(n)} \;\middle|\; \sigma^{(n)} \text{ bijetora}\right\}.$$

Seja $S_n^{j,i} \subset S_n$,

$$S_n^{j,i} \coloneqq \left\{\sigma \;\middle|\; \sigma(a_j) = a_i;\ j, i \text{ definidos}\right\}.$$

É óbvio que

$$\bigcup_i S_n^{j,i} = S_n, \qquad \forall j \in \{1, 2, \ldots, n\}.$$

Por simetria $\left|S_n^{j,i}\right| = \left|S_n^{j',i'}\right|$, $\forall\, i, j$.

Seja então

$$\lambda_{n,n+1} = \left\{x = (\sigma^{(n)}, \sigma^{(n+1)}) \;\middle|\; \sigma^{(n)} \cap \sigma^{(n+1)} = \sigma^{(n)}\right\}.$$

Note que $x \in \lambda_{n,n+1} \Rightarrow \sigma^{(n+1)}(a_{n+1}) = a_{n+1}$. Fixado $\sigma^{(n+1)}(a_{n+1}) = a_{n+1}$,

$$\begin{aligned}
&\implies \exists!\, \sigma^{(n+1)} \;\middle|\; \sigma^{(n+1)}(k) = \sigma^{(n)}(k), \quad \forall k,\ \sigma^{(n)} \\
&\implies \left|\lambda_{n,n+1}\right| = |S_n| = \left|S_{n+1}^{n+1,i}\right| \\
&\implies |S_{n+1}| = \left|\bigcup_i S_{n+1}^{n+1,i}\right| = \sum_i \left|S_{n+1}^{n+1,i}\right| = (n+1)\cdot|S_n|.
\end{aligned}$$

Como $|S_{n+1}| = (n+1)\cdot|S_n|$, por PIF $|S_1| = 1$

$$\iff |S_2| = 2 \iff \cdots \iff |S_n| = n! \qquad \text{C.Q.D.}$$